STAFF REVIEW · H446 099 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q2; Booklet Q3 = original online Q3

Do Now — source answers

1. Instruction Set Architecture—programmer-visible instruction operations.

2. Which operation the CPU performs.

3. Transfers values between registers and memory.

4. A∧(B∨¬B)=A.

I Do — source worked steps

1. Follow the reasoning — LOAD R1,A→5; LOAD R2,B→8; ADD R3,R1,R2→13; STORE C,R3→13.

We Do — guided CHECK

1. Follow the reasoning
R1=4,R2=7,R3=11, memory[C]=11.

You Do — source mark schemes

1. RISC regular/simple instructions, load/store architecture [2]; CISC richer/variable instructions, potentially combined work [2].

2. R1=6 [1], R2=9 [1], R3=15 [2], C=15 [2].

3. Both can pipeline [2]; RISC often needs more simple ISA instructions for complex task [2]; real microarchitecture and compiler/workload matter [2].

Exit Ticket — source answer

1. False. CISC processors use pipelines too.